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# Palaeomagnetism of Australian Hematite Ore Bodies—II Western Australia

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*Summary*

Iron ore deposits of the Lake Superior type within the Archaean and Proterozoic banded iron formations of Western Australia show consistent and stable directions of remanent magnetization. The palaeomagnetic poles of the ores do not correlate with poles of Post-Cambrian Australian rocks, which suggests a Pre-Cambrian age for the ore formation. Magnetic anisotropies of the ores are low and the distribution of the principal susceptibility axes tends to reflect the magnetic fabric of the host rocks.

## 1. Introduction

In a previous publication (Chamalaun & Porath 1967) we reported the results of a palaeomagnetic study on the hematite ore bodies of the Middleback Ranges in South Australia. It was found that these were consistently and stably magnetized with a Pre-Cambrian direction. We therefore extended the study to the Western Australian ore bodies, the results of which are reported here.

A large number of high-grade hematite ore bodies are known to occur in the Archaean and Proterozoic banded iron formations of Western Australia, but in order to obtain fresh subsurface samples we have sampled only those that are actively mined. The localities of the ore bodies sampled are shown in Fig. 1. The Mt Goldsworthy and Koolyanobbing deposits occur in Archaean banded iron formations, whilst the host rocks for the Mt Tom Price and Mt Newman ore bodies are Proterozoic iron formations.

## 2. Results

(a) *Mt Goldsworthy.* The Mt Goldsworthy deposit is located at the northern extremity of the Western Australian Archaean shield. A folded and faulted banded iron formation, which consists of alternating layers of silica and hematite, forms the host rock of a number of high-grade hematite ore bodies of varying origins. Brandt (1964, 1966) makes a distinction between lode ore and crust ore deposits. Lode ores are deep lenses of massive finely crystalline hematite (average grain size about 20$\mu$). The major deposit at Mt Goldsworthy consists of lode ore and is controlled by a steep angled transcurrent fault. Pebbles of massive hematite ore occur in the basal Proterozoic conglomerate indicating that the ore formed in Pre-Proterozoic times (Brandt 1964). Brandt (1966) favours a hypogene origin of the lode ore along the lines suggested by Dorr (1965) for the Brazilian ores.

Crust ores form shallow cappings of hematite and goethite on the upturned

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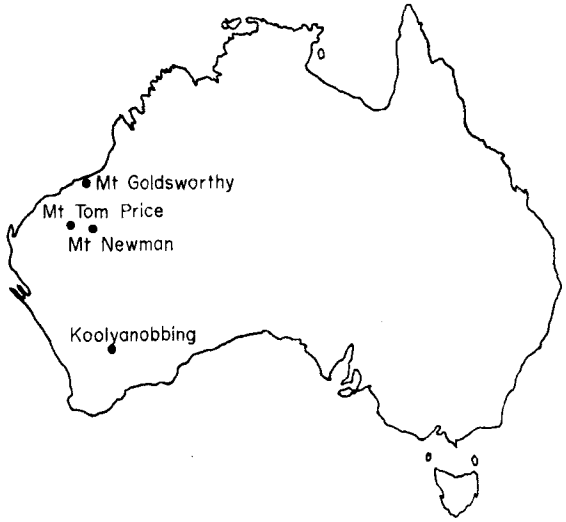



FIG. 1. Map of Australia showing the localities of the iron ore bodies sampled.

edges of the steeply dipping banded iron formation. The crust ores are found to overlie the lode ores in some places and are therefore younger. They have probably developed from the iron formation by supergene leaching of silica and substitution of silica by iron (Brandt 1964).

Sixty samples were collected from surface exposures and adits of the banded iron formation, lode ore and crust ore. The NRM directions, which are stable against partial demagnetization by thermal and alternating field methods, are shown in Fig. 2. Lode ores show a discrete blocking temperature with a Curie temperature of 695 °C, which is characteristic of hematite (see Fig. 3). The crust ores, however, show a spectrum which includes lower blocking temperatures, but their magnetization directions remain stable with remanent coercivities exceeding 750 oersteds.

A statistical test (Watson & Irving 1957) shows that the directions of magnetization of the lode ore do not have a Fisher (1953) distribution. ($\chi^2_{Dec} = 22{\cdot}9$; $\chi^2_{Inc} = 18{\cdot}3$.) The magnetization directions form two clusters, the majority (31) having moderate to steep inclinations and northwesterly declinations, whilst eight samples have steep inclinations and southeasterly declinations. Of these eight samples three come from a surface exposure of the ore body and another three from parts of the ore body which are near the fault zone and the contact zone with the banded iron formation.

Polar wandering is suggested as a possible explanation of the observed pattern of magnetization directions of the lode ore samples. The relative age of the two groups of directions is not known, but it is possible that the ore formation progressed from these zones inward, so that the minor group of directions represents the early stages in the ore-forming process. As each of the two clusters separately obeys the rules of Fisherian statistics (major group: $\chi^2_{Dec} = 9{\cdot}5$; $\chi^2_{Inc} = 9{\cdot}3$; minor group: $\chi^2_{Dec} = 10{\cdot}0$; $\chi^2_{Inc} = 2{\cdot}4$), the magnetization of the lode ore is represented by two mean directions and their corresponding palaeomagnetic poles (see Table 1).

The mean direction of the crust ore samples is significantly different from that of the lode ore. The circle of confidence of the mean direction is, however, rather large due to the fact that only a few samples could be collected from the extremely friable ore. Samples from the banded iron formation have magnetization directions similar to those of the main group of lode ore samples. The significance test of Watson & Irving (1957) shows that the two mean vectors are not significantly different.

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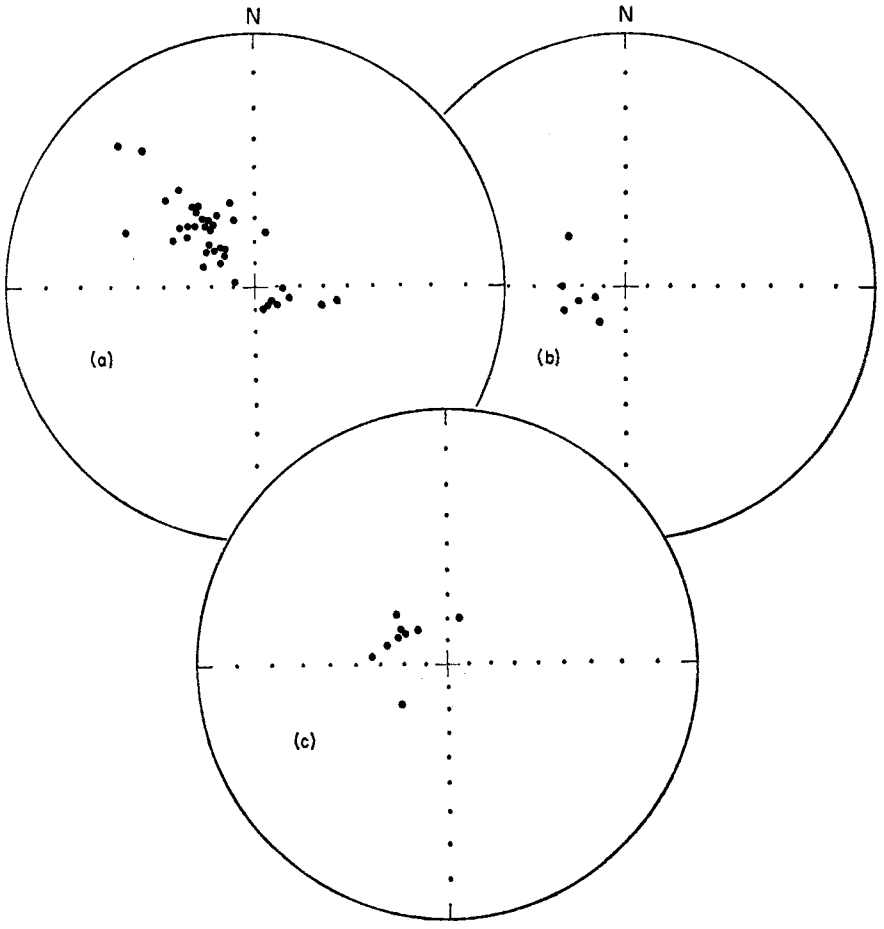



FIG. 2. Equal angle projections of the magnetization directions of the Mt Goldsworthy lode ores (a), crust ores (b) and banded iron formation (c). The symbols denote positive inclinations.

The magnetic anisotropies of the ore samples were determined with a low field torque meter (King & Rees 1962). Magnetic anisotropies are well below 1 per cent and thus would not have influenced the alignment of the magnetization vector along the ambient field direction during the formation of the ore. No consistent distribution of maximum and minimum susceptibility axes is obtained, the susceptibilities of the ores being low and rather uniform with a mean of

$$K = (58 \pm 2) \times 10^{-6} \text{ e.m.u./g},$$

reflecting the negligible content of magnetite and the small grain size of hematite.

(b) *Koolyanobbing*. A number of iron ore bodies occur as irregular lenses in the folded parts or along straight limbs of the extensively metamorphosed banded iron formation of the Koolyanobbing Hills of South West Australia. Dowd's Hill and the A-deposit are the largest ore bodies and have been sampled for magnetic studies. The mineralogy of the Koolyanobbing ores is much more complex compared with other Lake Superior type ores. Four types of iron oxides can be distinguished in the Dowd's Hill deposit. Cryptocrystalline goethite is predominant and always accompanied by varying amounts of magnetite and hematite in the form of martite.

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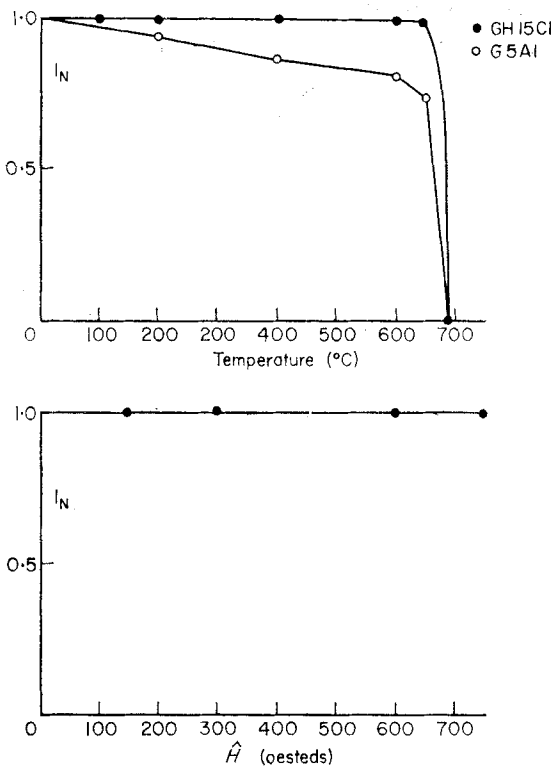



FIG. 3. Normalized thermal (top) and alternating field (bottom) demagnetization curves for hematite ore specimens from the Mt Goldsworthy deposit. GH15C1 is a specimen of lode ore, G5A1 a specimen of crust ore.

## Table 1

*Mean directions, palaeomagnetic poles and precision parameters for the Mt Goldsworthy hematite ores*

| | | Mean direction | | | | | Pole | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Formation | $N$ | $D$ | $I$ | $R$ | $K$ | $\alpha_{95}$ | Lat. | Long. | $K$ | $\alpha_{95}$ |
| Lode Ore G1 | 31 | 317° | 54½° | 29·672 | 22·6 | 5½° | 19½° N | 84° E | 19 | 6 |
| Lode Ore G3 | 8 | 115° | 73½° | 7·843 | 45 | 8½° | 30½° S | 149½° E | 20 | 12½ |
| Crust Ore G2 | 6 | 263° | 68½° | 5·838 | 31 | 12° | 21½° S | 79° E | 13 | 19° |
| Banded iron formation | 9 | 298° | 68° | 8·698 | 27 | 10° | — | — | — | — |

A later stage of hydrothermal activity has been responsible for the development of a second generation of hematite in the form of coarsely crystalline and specular hematite (Miles 1953). No specularite veins are evident in the A-deposit. Here the ore consists of goethite with varying amounts of magnetite and martite as well as patches of fine grained earthy hematite.

Although the mineralogy of the ore is rather complex compared with other Lake Superior type ores, surprisingly well grouped directions of magnetization are obtained for the Dowd's Hill deposit (Fig. 4(a), (b)). Partial demagnetization by alternating field and thermal methods results in some improvement of the original grouping. The mean direction of magnetization, palaeomagnetic pole and precision parameters are listed in Table 2. The magnetization directions follow a Fisher distribution, as the $\chi^2$-parameters are not significant ($\chi^2_{Dec} = 6{\cdot}4$; $\chi^2_{Inc} = 7{\cdot}0$).

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## Table 2

*Mean direction, palaeomagnetic poles and precision parameters for the Dowd's Hill ore body, Koolyanobbing*

| | Mean direction | | | | | Pole | | | |
|---|---|---|---|---|---|---|---|---|---|
| $N$ | $D$ | $I$ | $R$ | $K$ | $\alpha_{95}$ | Lat. | Long. | $K$ | $\alpha_{95}$ |
| 35 | 303° | −60½° | 32·704 | 15 | 6½° | 43° N | 4° W | 9 | 8½° |

Thermal and alternating field decay curves show a variety of blocking temperatures and remanent coercivities. The NRM contains both magnetite and hematite Curie points as evident from the thermal demagnetization curves in Fig. 5. In general, a distributed spectrum of blocking temperatures is observed, except for the coarse crystalline and specular hematite samples, which show a discrete blocking temperature. No change in the direction of magnetization is observed after heating above 600° with the exception of those samples where magnetite is preponderant and acquires a TRM in stray fields in the furnace. This demonstrates that magnetite

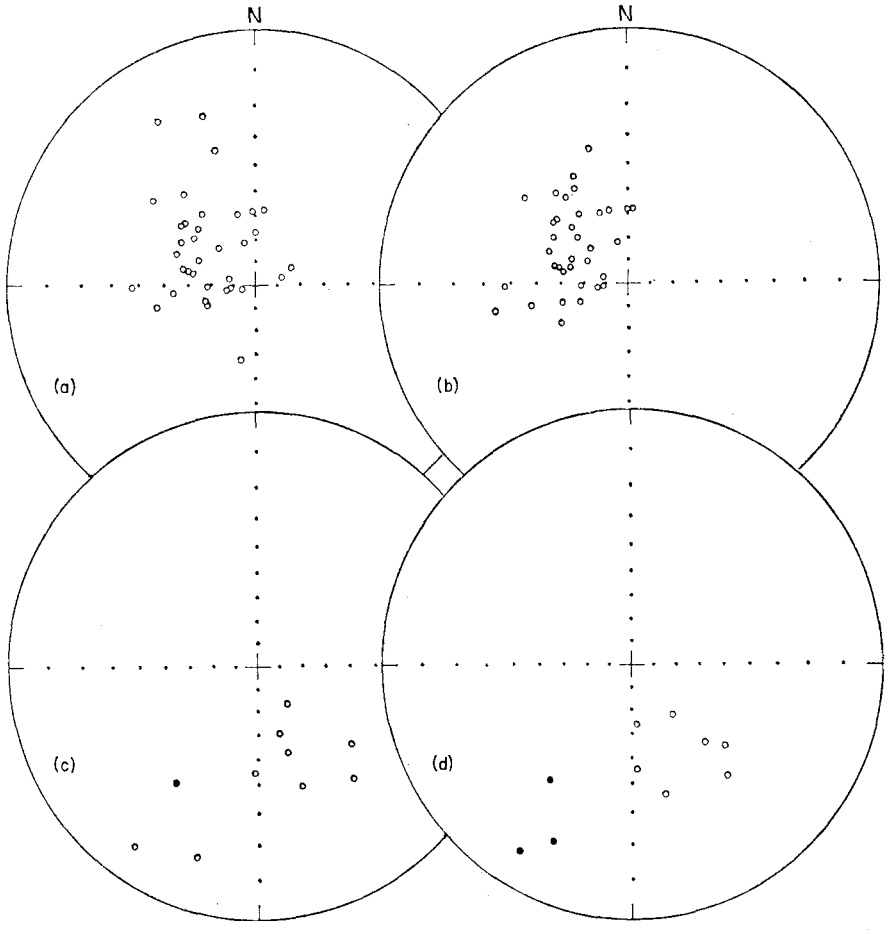



FIG. 4. Equal angle projections of the magnetization directions of the Koolyanobbing ore bodies, Dowd's Hill (a)–(b) and the A-deposit (c)–(d), before (left) and after (right) magnetic cleaning. Open symbols denote negative inclinations; solid symbols denote positive inclinations.

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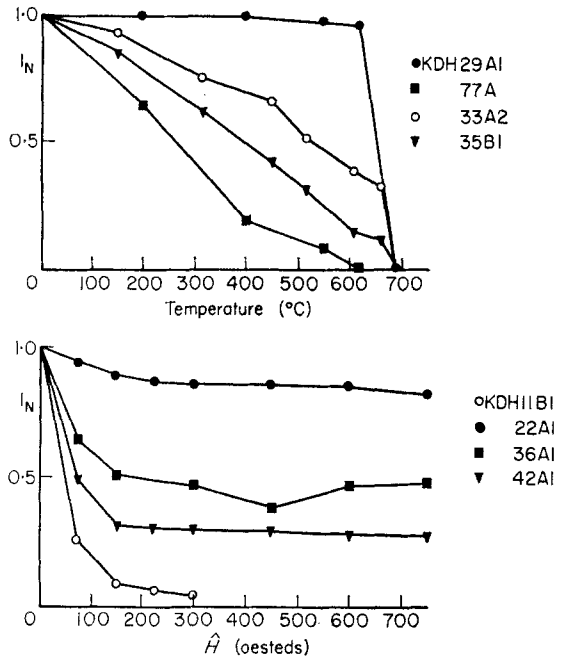



FIG. 5. Normalized thermal (top) and alternating field (bottom) demagnetization curves of iron ore specimens from the Dowd's Hill deposit. KDH29A1 and KDH22A1 are specimens of coarse crystalline hematite ore.

and hematite have the same direction of magnetization, which in turn is the same as the NRM directions of the coarse crystalline and specular hematite samples. It appears therefore that the magnetization of this ore body is related to the formation of the specular hematite, which represents the latest stage in the development of iron oxides.

Alternating field demagnetization was preferred for the final cleaning of the collection, as a large number of specimens of geothite ore exploded during thermal demagnetization at about 300 °C, the dehydration temperature of goethite. The NRM shows a variety of remanent coercivities (Fig. 5). Samples with low remanent coercivities move towards the centre of the group, but generally changes in the magnetization directions are only small. Magnetization directions for samples from the A-deposit do not form a tight group (Fig. 4(c), (d)). Samples are too scarce and the dispersion of the directions too large to justify the calculation of a mean direction.

Magnetic anisotropies of the ores are substantially higher than the high-grade hematite deposits. Magnetic lineations and foliations are several per cent for the Dowd's Hill deposit and reach up to 15 per cent for samples from the A-deposit. Maximum susceptibility axes of the Dowd's Hill ore body tend to lie within the plane of banding of the host rocks, but minimum axes diverge significantly from the pole of the banding plane (Fig. 6). Susceptibilities of the ores are very variable due to the different magnetite contents.

(c) *Hamersley Iron Province.* The Hamersley Province in North-West Australia is one of the largest iron ore provinces of the world. Concentrations of iron ore are confined to the base member of the Brockman Iron Formation, which belongs to the Hamersley Group, a succession of predominantly chemical sediments. A member of the acid volcanics within the Hamersley Group has been dated at 2100 m.y. (Leggo *et al.* 1965). Mt Tom Price and Mt Newman represent the largest deposits and have been sampled for palaeomagnetic measurements. Both ore bodies

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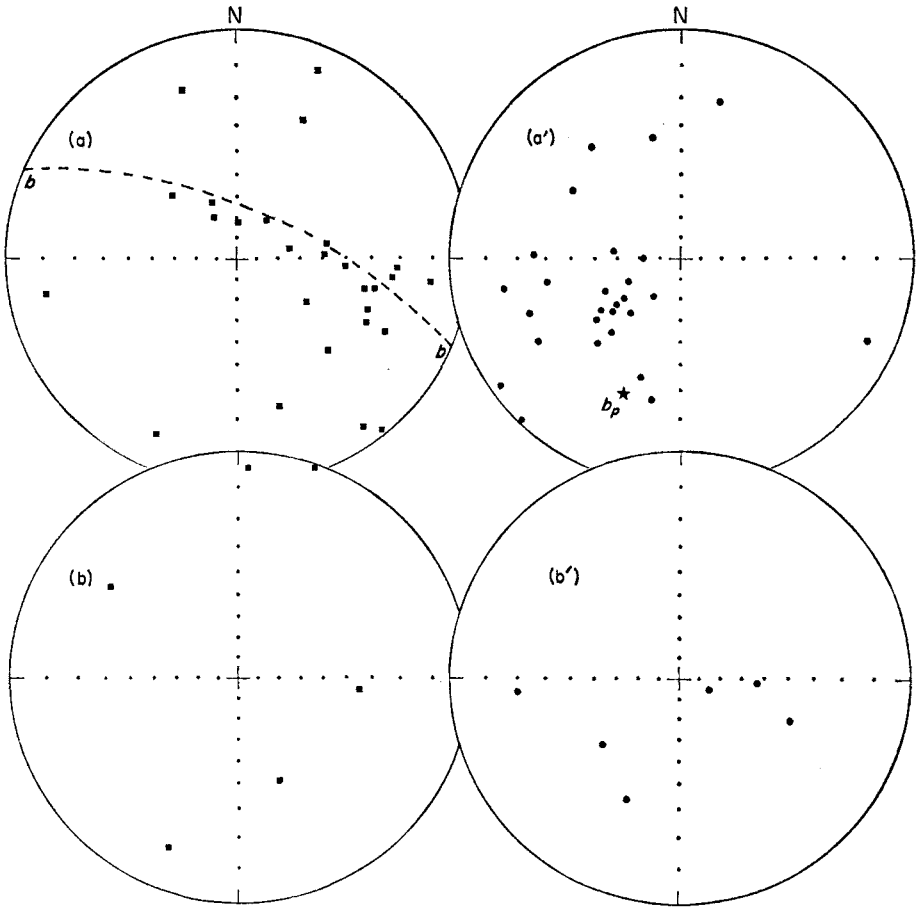



FIG. 6. Lower hemisphere equal angle projections of maximum (left) and minimum (right) susceptibility axes for the Dowd's Hill (a)–(a') and the A-deposit (b)–(b'). $b$–$b$ denotes the banding plane and $b_p$ the pole of the banding plane of the ore.

consist of massive finely crystalline hematite. The banding of the protore beds is well preserved.

Iron ore concentrations in the Hamersley Province are found to be related to an old land surface which is suggested by McLeod (1963) to date from Mesozoic times. McLeod suggests, therefore, that the age of the ore formation is relatively recent compared with the majority of the Lake Superior type ores, which are Pre-Cambrian in age.

Sixty samples were collected from the Brockman Iron Formation and the hematite ore bodies, Mt Tom Price and Mt Newman. The NRM directions of the Brockman Iron Formation are widely scattered and the dispersion does not decrease after partial demagnetization by alternating field and thermal methods. The remanent magnetization is carried by magnetite, which shows very low remanent coercivities (10–80 oersteds). Samples have predominantly low inclinations with respect to the banding, which might be due to the strong textural anisotropy of the iron formation.

The NRM directions of samples from the Mt Newman and Mt Tom Price ore bodies are shown in Fig. 7. Mt Newman samples carry a consistent and stable remanent magnetization; the demagnetization curves showing a discrete blocking temperature and remanent coercivities exceeding 750 oersteds. On the other hand, the NRM directions of the Mt Tom Price samples are scattered, but the dispersion

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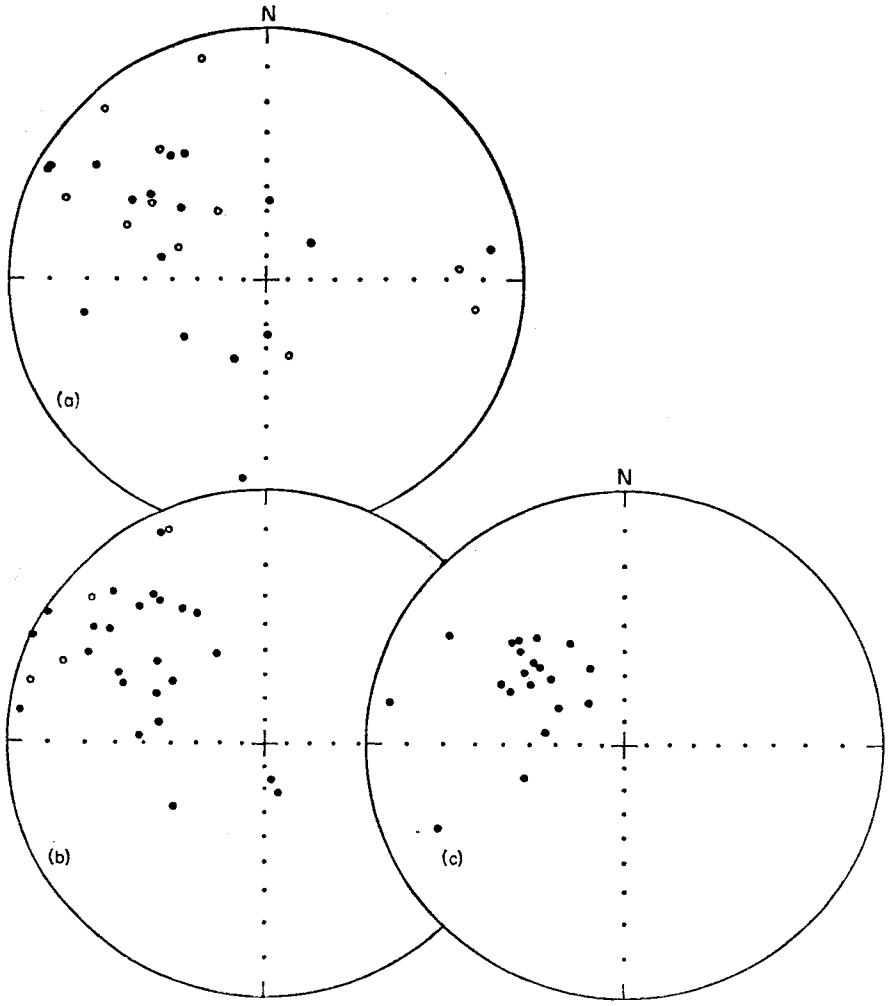



FIG. 7. Equal angle projections of the magnetization directions of samples from the Mt Tom Price and Mt Newman ore bodies. (a) Mt Tom Price before magnetic cleaning; (b) Mt Tom Price after magnetic cleaning; (c) Mt Newman.

decreases considerably after thermal and alternating field treatment in 600° and 750 oersteds, respectively. Some demagnetization curves are shown in Fig. 8. Low blocking temperatures and low remanent coercivities are generally due to the small fraction of residual magnetite.

Mean directions of magnetization, palaeomagnetic poles and precision parameters of Mt Newman and Mt Tom Price ore bodies are listed in Table 3. Both groups of magnetization directions follow the rules of the Fisherian statistics (Mt Tom Price: $\chi^2_{Dec} = 5{\cdot}1$, $\chi^2_{Inc} = 5{\cdot}7$; Mt Newman: $\chi^2_{Dec} = 6{\cdot}4$, $\chi^2_{Inc} = 7{\cdot}8$). The mean directions are similar and the circles of confidence overlap. The dispersions of magnetization directions of the two ore bodies are, however, significantly different so that it is not possible to test for the identity of the two mean vectors (Watson & Irving 1957).

The distribution of the maximum and minimum susceptibility axes of Mt Newman and Mt Tom Price ore bodies with respect to the banding plane are shown in Fig. 9. Minimum susceptibility axes tend to scatter around the normal of the banding plane, whereas maxima lie mainly within the banding plane. This reflects the magnetic fabric of the protore beds which is caused by alternating layers of silica and iron

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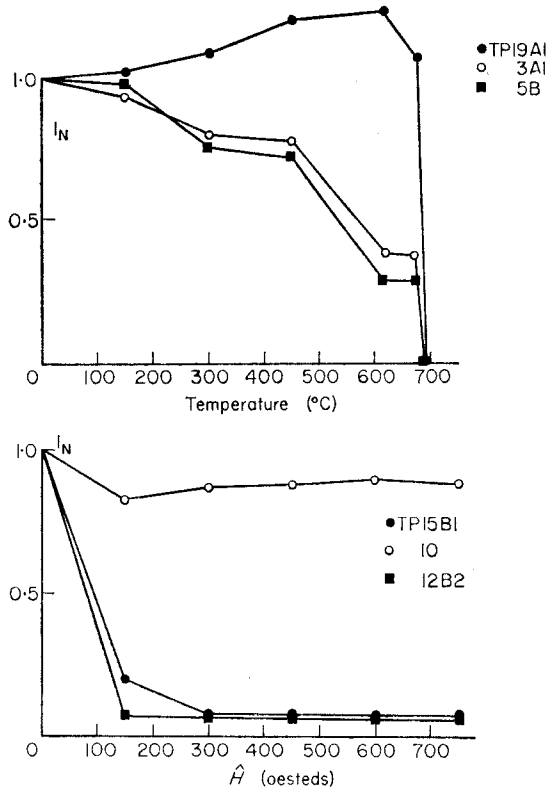



FIG. 8. Normalized thermal (top) and alternating field (bottom) demagnetization curves for specimens from the Mt Tom Price ore body.

## Table 3

*Mean directions, palaeomagnetic poles and precision parameters of the Hamersley ore bodies, Mt Tom Price and Mt Newman*

| | | Mean direction | | | | | Pole | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Ore body | $N$ | $D$ | $I$ | $R$ | $K$ | $\alpha_{95}$ | Lat. | Long. | $K$ | $\alpha_{95}$ |
| Mt Tom Price | 28 | 304° | 25° | 23·588 | 6·1 | 12° | 22° N | 57° E | 6 | 12° |
| Mt Newman | 20 | 302° | 39° | 18·241 | 12 | 9½° | 17° N | 66° E | 12 | 9½° |

oxide. It is possible that bands in the hematite ore that have developed from the magnetite bands of the banded iron formation contain a larger amount of residual magnetite than those bands that have originated from the silica bands by substitution of iron, thus causing a textural anisotropy of the hematite ore. The anisotropies are again only very small and magnetic lineations and foliations generally well below 1 per cent. Susceptibilities are of the order of $100 \times 10^{-6}$ e.m.u./g and somewhat larger for samples from the Mt Tom Price deposit, probably because of a higher content of residual magnetite.

### 3. Magnetic properties of the ores

The high-grade hematite ores of Mt Goldsworthy, Mt Newman and Mt Tom Price show the characteristic magnetic properties of fine grained hematite, i.e. a discrete blocking temperature and high coercivities. Similarly the low temperature transi-

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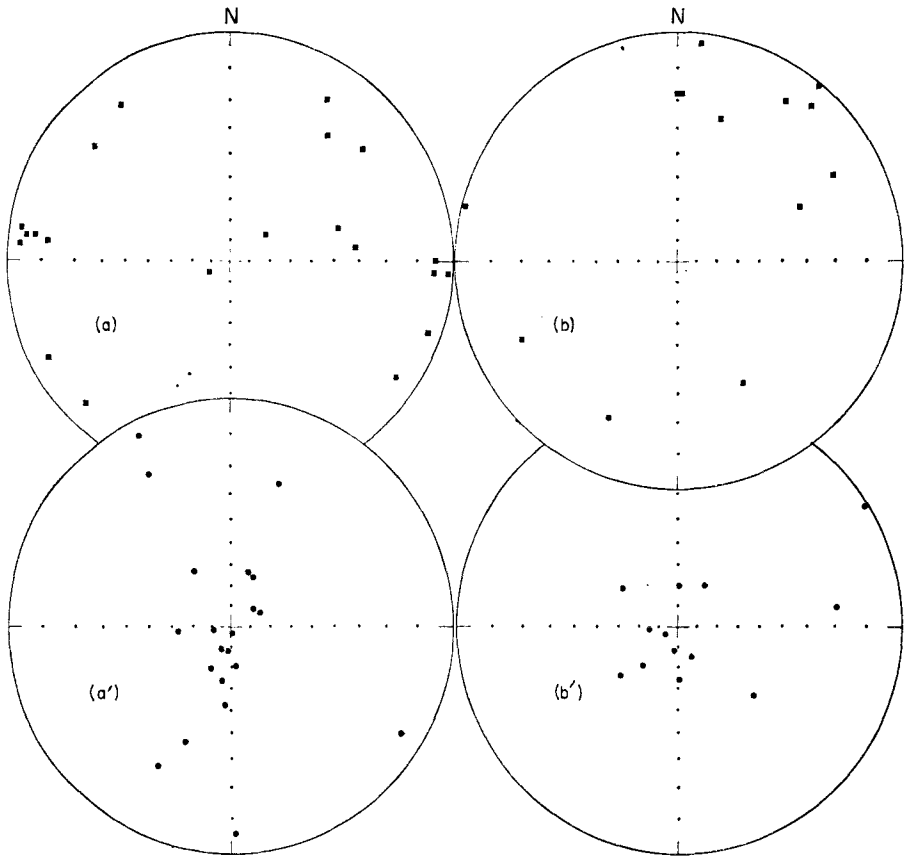



FIG. 9. Lower hemisphere equal angle projections of the maximum (top) and minimum (bottom) susceptibility axes for samples from the Mt Tom Price (a)–(a′) and Mt Newman (b)–(b′) ore bodies with respect to the banding plane.

tion of hematite (Morin 1950) is observed for all hematite ores. Only about 10 per cent of the NRM intensity is retained at −80 °C and between 40 and 60 per cent of the original magnetization is recovered after re-heating to room temperature in zero field. The magnetization directions do not change during the low temperature treatment.

The NRM intensities of the ores are remarkably low and represent only 1 per cent or less of the TRM developed by cooling in a magnetic field of $H = 0{\cdot}6$ oersteds. Using the same reasoning as in the previous paper, we suggest that a CRM rather than a TRM is responsible for the natural remanence of the ores.

## 4. Discussion of results

The Lake Superior ore bodies reported here carry a consistent and stable direction of remanent magnetization. Magnetic anisotropies are low and will not have seriously affected the direction of the magnetization vector. By comparing the palaeomagnetic poles of the ores with the Australian polar wandering curve it should then be possible to estimate the age of the ore formation. Reliable palaeomagnetic poles have, however, only been obtained from Post-Cambrian Australian rocks and the poles for the iron ores, which are shown in Fig. 10, do not correlate with the established Australian polar wandering curve. This suggests a Pre-Cambrian age of

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the ore formation. A comparison with preliminary results on Australian Pre-Cambrian rocks (Irving & Green 1958) shows the palaeomagnetic pole of the Dowd's Hill deposit to be in good agreement with the pole of the Nullagine Lavas, if sign is disregarded (see Fig. 10). Although the exact sampling sites have not been described by Irving & Green, the Nullagine Lavas probably form part of the Lower Proterozoic Fortescue Group which is older than 2100 m.y. (Leggo *et al.* 1965). A lower Proterozoic age would therefore result for the formation of the largest Koolyanobbing ore body.

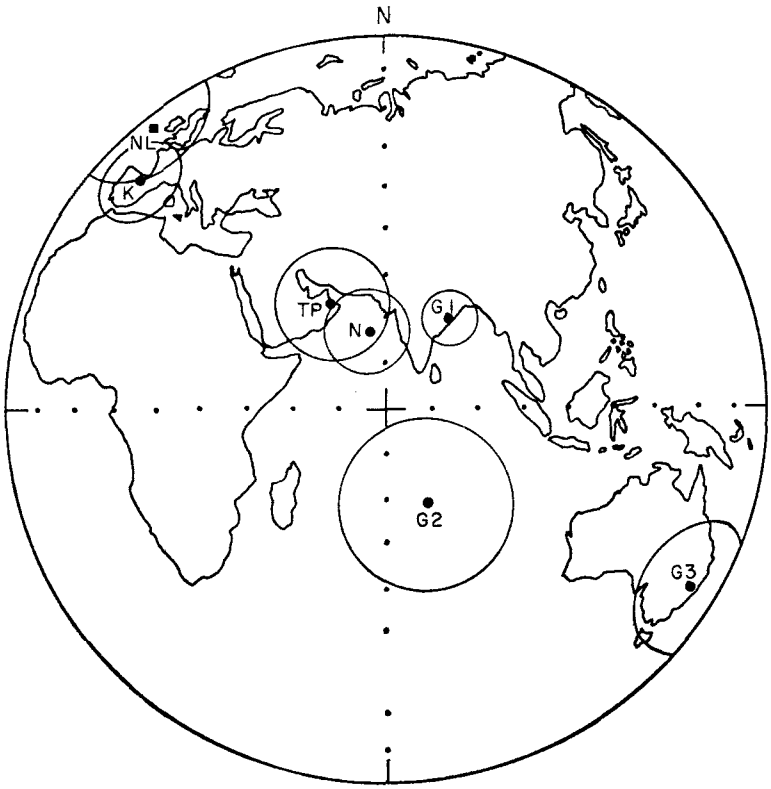



FIG. 10. Equal area projection of the palaeomagnetic pole positions of the Western Australian iron ore bodies; G1, G3 = Mt Goldsworthy lode ore; G2 = Mt Goldsworthy crust ore; TP = Mt Tom Price; N = Mt Newman; K = Dowd's Hill. NL denotes the pole of the Nullagine Lavas from Irving & Green (1958).

Because of the lack of Australian Pre-Cambrian palaeomagnetic poles equivalent to the poles of the Mt Goldsworthy and Hamersley deposits, it is not possible to arrive at a more accurate age for the formation of these ores. The distribution of the magnetization directions of the Mt Goldsworthy deposits suggests that the ore formation covered a considerable period of time. The difference in age between the lode ore and the crust ore, which was suggested by Brandt (1964) on geological evidence, is confirmed by the magnetic results. Samples from the banded iron formation have similar magnetization directions to the ores. The iron oxide of the Archaean banded iron formation at Mt Goldsworthy is formed by hematite in similar crystalline form as in the hematite ore bodies. It is therefore likely that the remanence of the banded iron formation dates from the formation of the ore and that the hematite in the iron formation signifies the first phase of the formation of an ore body. The

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fact that pebbles of massive hematite ore have been found in the basal Proterozoic conglomerate suggests that the palaeomagnetic poles of the Mt Goldsworthy lode ore deposit represent early Pre-Cambrian poles.

The position of the palaeomagnetic poles of the Hamersley ore bodies, Mt Tom Price and Mt Newman, does not support a Mesozoic age of the ore formation as was suggested by McLeod (1963), but indicates that, similar to other Lake Superior type ores, they are Pre-Cambrian in age. It appears therefore that the old land surface, to which concentrations of iron ore are confiend in the Hamersley Province, dates from Pre-Cambrian times. To assign a more accurate age for the ore formation fails again because of the lack of palaeomagnetic data from Australian Pre-Cambrian rocks.

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